



UNIVERSITY TECHNOLOGY MALAYSIA

**LAB EXERCISE 2
SECJ1013-03
(PROGRAMMING TECHNIQUE I)**

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Code :

```
#include <iostream>
#include <iomanip>
#include <cmath>
using namespace std;

//Prototypes
void matrices();
double euclidean(int, int, int, int);

// Global Variables
int x_1 = 1, y_1 = 3, x_2 = 2, y_2 = 6, x_3 = 5, y_3 = 4;

int main()
{
    // Constructing the table
    cout << "\n";
    matrices();
    cout << "\n";

    // Distance AB
    double euclidean (int x_1, int x_2, int y_1, int y_2);
    cout << "\t" << "AB = " << euclidean(x_1, x_2, y_1, y_2) << "\n";

    // Distance AC
    double euclidean (int x_1, int x_3, int y_1, int y_3);
    cout << "\t" << "AC = " << euclidean(x_1, x_3, y_1, y_3) << "\n";

    // Distance BC
    double euclidean (int x_2, int x_3, int y_2, int y_3);
    cout << "\t" << "BC = " << euclidean(x_2, x_3, y_2, y_3) << "\n";

    return 0;
}

void matrices()
{
    int x [3] = {1, 2, 5};
    int y [3] = {3, 6, 4};
    char coordinate [3][3] = {"A", "B", "C"};

    for (int count = 0; count < 3; count++)
    {
        if (count == 0)
```

```

        {
            cout << "\t" << coordinate[count] << "(" << x[count] << "," <<
y[count] << ")";
        }
        else if (count == 2 )
        {
            cout << ", and " << coordinate[count] << "(" << x[count] << "," <<
y[count] << ")";
        }
        else
        {
            cout << ", " << coordinate[count] << "(" << x[count] << "," <<
y[count] << ")";
        }
    }

    cout << "\n";
    cout << "\n";

    cout << "\t" << " " << setw(9) << "x" << setw(9) << "y" << "\n";

    for (int count = 0; count < 3; count++)
    {
        cout << "\t" << coordinate[count] << setw(9) << x[count] << setw(9) <<
y[count] << "\n";
    }

}

double euclidean(int a, int b, int c, int d)
{
    // Difference of both x and y
    int difference_x = a - b;
    int difference_y = c - d;

    // The square of the difference of x and y
    int xpower2 = pow(difference_x,2);
    int ypower2 = pow(difference_y,2);

    // Square root of the sum of squares
    double distance = sqrt(xpower2 + ypower2);

    return distance;
}

```

```

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2 #include <iomanip>
3 #include <cmath>
4 using namespace std;
5
6 //Prototypes
7 void matrices();
8 double euclidean(int, int, int, int);
9
10 // Global Variables
11 int x_1 = 1, y_1 = 3, x_2 = 2, y_2 = 6, x_3 = 5, y_3 = 4;
12
13 int main()
14 {
15     // Constructing the table
16     cout << "\n";
17     matrices();
18     cout << "\n";
19
20     // Distance AB
21     double euclidean (int x_1, int x_2, int y_1, int y_2);
22     cout << "\t" << "AB = " << euclidean(x_1, x_2, y_1, y_2) << "\n";
23
24     // Distance AC
25     double euclidean (int x_1, int x_3, int y_1, int y_3);
26     cout << "\t" << "AC = " << euclidean(x_1, x_3, y_1, y_3) << "\n";
27
28     // Distance BC
29     double euclidean (int x_2, int x_3, int y_2, int y_3);
30     cout << "\t" << "BC = " << euclidean(x_2, x_3, y_2, y_3) << "\n";
31
32     return 0;
33 }
34
35 void matrices()
36 {
37     int x [3] = {1, 2, 5};
38     int y [3] = {3, 6, 4};
39     char coordinate [3][3] = {"A", "B", "C"};
40
41     for (int count = 0; count < 3; count++)
42     {
43         if (count == 0)
44         {
45             cout << "\t" << coordinate[count] << "(" << x[count] << "," << y[count] << ")";
46         }
47         else if (count == 2)
48         {
49             cout << ", and " << coordinate[count] << "(" << x[count] << "," << y[count] << ")";
50         }
51         else
52         {
53             cout << ", " << coordinate[count] << "(" << x[count] << "," << y[count] << ")";
54         }
55     }
56
57     cout << "\n";
58     cout << "\n";
59
60     cout << "\t" << " " << setw(9) << "x" << setw(9) << "y" << "\n";
61
62     for (int count = 0; count < 3; count++)
63     {
64         cout << "\t" << coordinate[count] << setw(9) << x[count] << setw(9) << y[count] << "\n";
65     }
66 }
67
68 double euclidean(int a, int b, int c, int d)
69 {
70     // Difference of both x and y
71     int difference_x = a - b;
72     int difference_y = c - d;
73
74     // The square of the difference of x and y
75     int xpower2 = pow(difference_x,2);
76     int ypower2 = pow(difference_y,2);
77
78     // Square root of the sum of squares
79     double distance = sqrt(xpower2 + ypower2);
80
81     return distance;
82 }
83

```

Output:

```
#include <iostream>
#include <iomanip>
#include <cmath>
using namespace std;

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// Global Variables
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    cout << "\t" << "AB = " << euclidean(x_1, x_2, y_1, y_2);

    // Distance AC
    double euclidean(int x_1, int x_3, int y_1, int y_3)
    cout << "\t" << "AC = " << euclidean(x_1, x_3, y_1, y_3);

    // Distance BC
    double euclidean(int x_2, int x_3, int y_2, int y_3)
    cout << "\t" << "BC = " << euclidean(x_2, x_3, y_2, y_3);

    return 0;
}
```

A(1,3), B(2,6), and C(5,4)

	x	y
A	1	3
B	2	6
C	5	4

AB = 3.16228
AC = 4.12311
BC = 3.06555

Process exited after 0.05496 seconds with return value 0
Press any key to continue . . .

Compilation results...

Errors: 0
Warnings: 0
Output Filename: C:\Users\User\OneDrive\Documents\Education\UTM\CLASSES\SEM 1\SECJ1013-PROGRAMMING TECHNIQUE I-03\LAB\LAB 2\CODS\Euclidean Theory.exe
Output Size: 1.86186122894287 MiB
Compilation Time: 0.49s